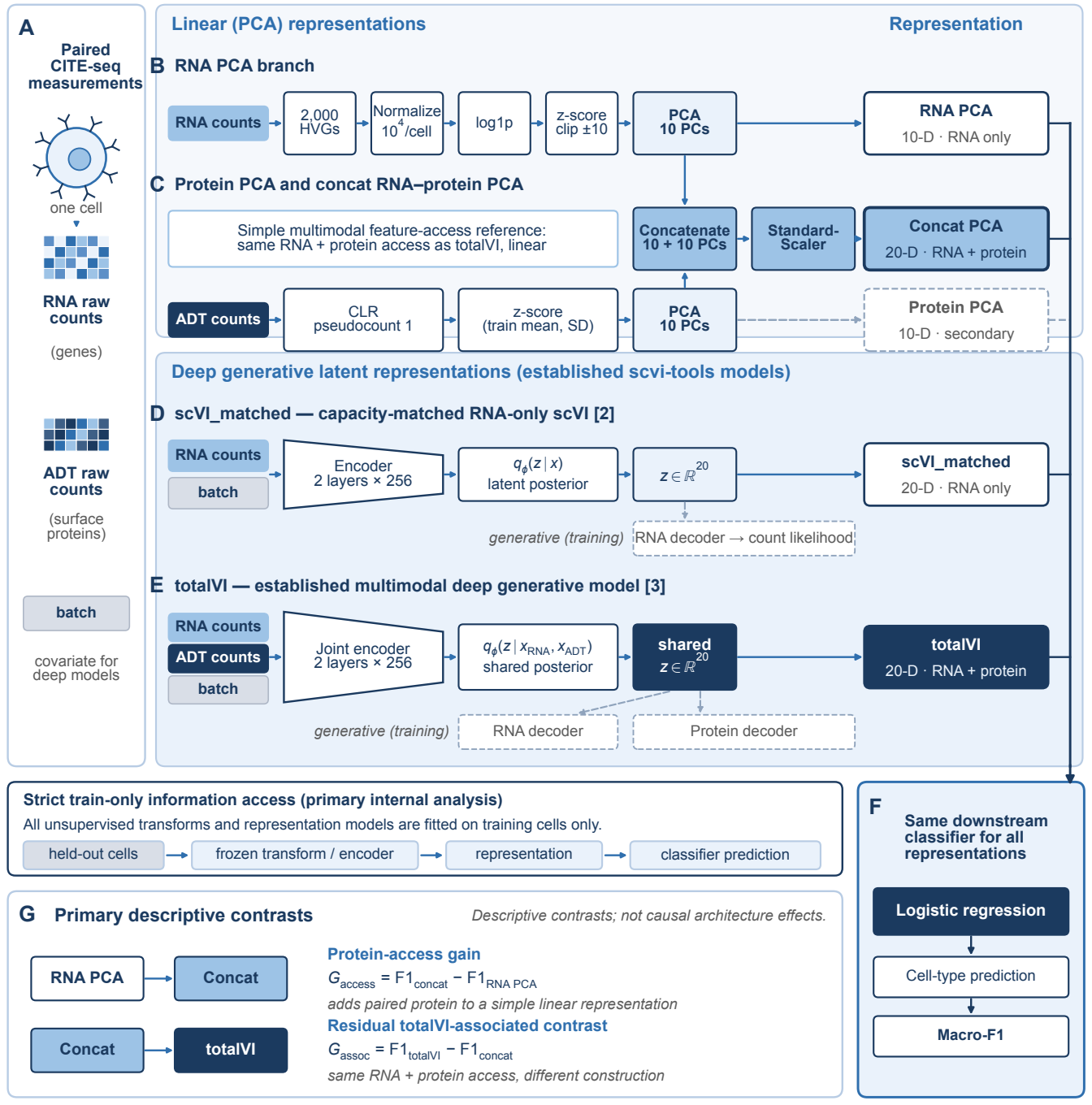

A
Paired CITE-seq measurements
one cell
RNA raw counts
(genes)
ADT raw counts
(surface proteins)
batch
covariate for deep models
Linear (PCA) representations
Representation
B RNA PCA branch
RNA counts
2,000 HVGs
Normalize 10^4/cell
log1p
z-score clip ±10
PCA 10 PCs
RNA PCA
10-D · RNA only
C Protein PCA and concat RNA–protein PCA
Simple multimodal feature-access reference: same RNA + protein access as totalVI, linear
Concatenate 10 + 10 PCs
Standard-Scaler
Concat PCA
20-D · RNA + protein
ADT counts
CLR pseudocount 1
z-score (train mean, SD)
PCA 10 PCs
Protein PCA
10-D · secondary
Deep generative latent representations (established scvi-tools models)
D scVI_matched — capacity-matched RNA-only scVI [2]
RNA counts
batch
Encoder 2 layers × 256
$q_\phi(z \mid x)$ latent posterior
$z \in \mathbb{R}^{20}$
scVI_matched
20-D · RNA only
generative (training)
RNA decoder → count likelihood
E totalVI — established multimodal deep generative model [3]
RNA counts
ADT counts
batch
Joint encoder 2 layers × 256
$q_\phi(z \mid x_{RNA}, x_{ADT})$ shared posterior
shared $z \in \mathbb{R}^{20}$
totalVI
20-D · RNA + protein
generative (training)
RNA decoder
Protein decoder
Strict train-only information access (primary internal analysis)
All unsupervised transforms and representation models are fitted on training cells only.
held-out cells
frozen transform / encoder
representation
classifier prediction
F
Same downstream classifier for all representations
Logistic regression
Cell-type prediction
Macro-F1
G Primary descriptive contrasts
Descriptive contrasts; not causal architecture effects.
RNA PCA
Concat
Concat
totalVI
Protein-access gain
$G_{access} = F1_{concat} - F1_{RNA\ PCA}$
adds paired protein to a simple linear representation
Residual totalVI-associated contrast
$G_{assoc} = F1_{totalVI} - F1_{concat}$
same RNA + protein access, different construction